

Jiguang Li

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Education

- **The University of Chicago Booth School of Business** **Chicago, IL Sep 2022 – Jun 2027**
 - Ph.D. student in Econometrics and Statistics and M.B.A. candidate, supervised by *Veronika Ročková*
 - Research interests: Bayesian statistics, reinforcement learning, psychometrics
 - **Yale University** **New Haven, CT Aug 2019 – May 2020**
 - Master of Arts in Statistics
 - **Middlebury College** **Middlebury, VT Sep 2015 – May 2019**
 - Bachelor of Arts in Mathematics and Bachelor of Arts in Computer Science
 - Summa Cum Laude, Highest Honors in Mathematics, Davis UWC Scholar
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Publications

- **Jiguang Li**, Sid Kankanala, and Veronika Ročková. "Empirical Likelihood with Generative AI." Submitted, 2026. [paper link][code link]
 - Bengusu Nar, **Jiguang Li**, Veronika Ročková, and Panos Toulis. "Dynamic Treatment on Networks." Submitted, 2026. [paper link]
 - **Jiguang Li** and Henry Luo. "Robust Bayesian Optimization via Tempered Posteriors." Submitted, 2026. [paper link][code link]
 - **Jiguang Li**, Robert Gibbons, and Veronika Ročková. "Deep Computerized Adaptive Testing." Accepted for publication in *Psychometrika*, 2026. [paper link][code link]
 - **Jiguang Li**, Robert Gibbons, and Veronika Ročková. "Sparse Bayesian Multidimensional Item Response Theory." *Journal of the American Statistical Association*, 2025. [paper link][code link]
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Core Skills

Languages: Chinese (native), English (fluent), Spanish (college-level intermediate), Italian (IB ab initio)

Programming Languages: Python (PyTorch), R, Java, $\LaTeX$, basic JavaScript, HTML, and C

Experience

- **Applied Scientist Intern** **Amazon.com, Inc.**
Grocery Economics & Optimization Science *Jun 2025 – Aug 2025*
 - Built a constrained optimizer for media-channel budget allocation using a Bayesian marketing mix model for Whole Foods and Amazon Fresh, projecting an 8% ROAS lift with uncertainty quantification.
 - Built interactive visualization tools to make the optimization workflow interpretable for stakeholders.
- **Full-Time Research Professional, Center for Applied AI** **The University of Chicago**
Supervised by Sendhil Mullainathan *Aug 2020 – Jul 2022*
 - Developed PyTorch pipelines for medical imaging (CNNs, MAML, multi-GPU) and StyleGAN2.
 - Built Bayesian inference workflows in PyMC3/HMC for educational testing; analyzed large-scale item-response, DICOM X-ray, and conviction datasets.

- **Summer Research Assistant on Astrostatistics**

Supervised by George Djorgovski and Eilat Glikman

California Institute of Technology

Summers 2017–2019

- Implemented spectral continuum normalization methods in Python.
 - Built a quasar variability analysis pipeline and implemented two-sample tests; received the Caltech VURP award in 2017.
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Professional Service

- **Journal Reviewer:** *Bayesian Analysis, Operations Research, Psychometrika, Statistics and Computing.*
 - **Conference Reviewer:** NeurIPS.
 - **Teaching Assistant:** Business Statistics (BUSN41000, Booth), Linear Algebra (MATH0200, Middlebury), Macro Theory (ECON0250, Middlebury).
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Presentations

- Contributed talk, J-ISBA Conference (Jun 2024, Venice, Italy)
 - Contributed poster, ISBA Conference (Jul 2024, Venice, Italy)
 - Invited talk, Chicago Booth Healthcare Initiatives (Nov 2024, Chicago, IL)
 - Invited talk, Department of Public Health Sciences at UChicago (Oct 2025, Chicago, IL)
 - Invited talk, ICSA Conference (Jun 2026, Arlington, VA)
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Selected Coursework

- **Economics:** Bayesian Econometrics, Causal Machine Learning, Empirical Analysis, Price Theory 2 & 3, Theory of Income 1.
- **Mathematics:** Advanced Probability, Measure Theory, Optimization Techniques, Spectral Graph Theory.
- **MBA:** Financial Accounting, Marketing Data Science, Managing in Organizations, Investments, Operations Management.
- **Statistics/ML:** Bayesian Deep Learning, Fundamentals of Deep Learning, High-dimensional Hypothesis Testing, Linear Models, Mathematical Statistics 2.